

# ASSIGNMENT 6.3

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BATCH – 01

HALLTICKET NO. : 2303A51017

## Task Description #1 (Loops – Automorphic Numbers in a Range)

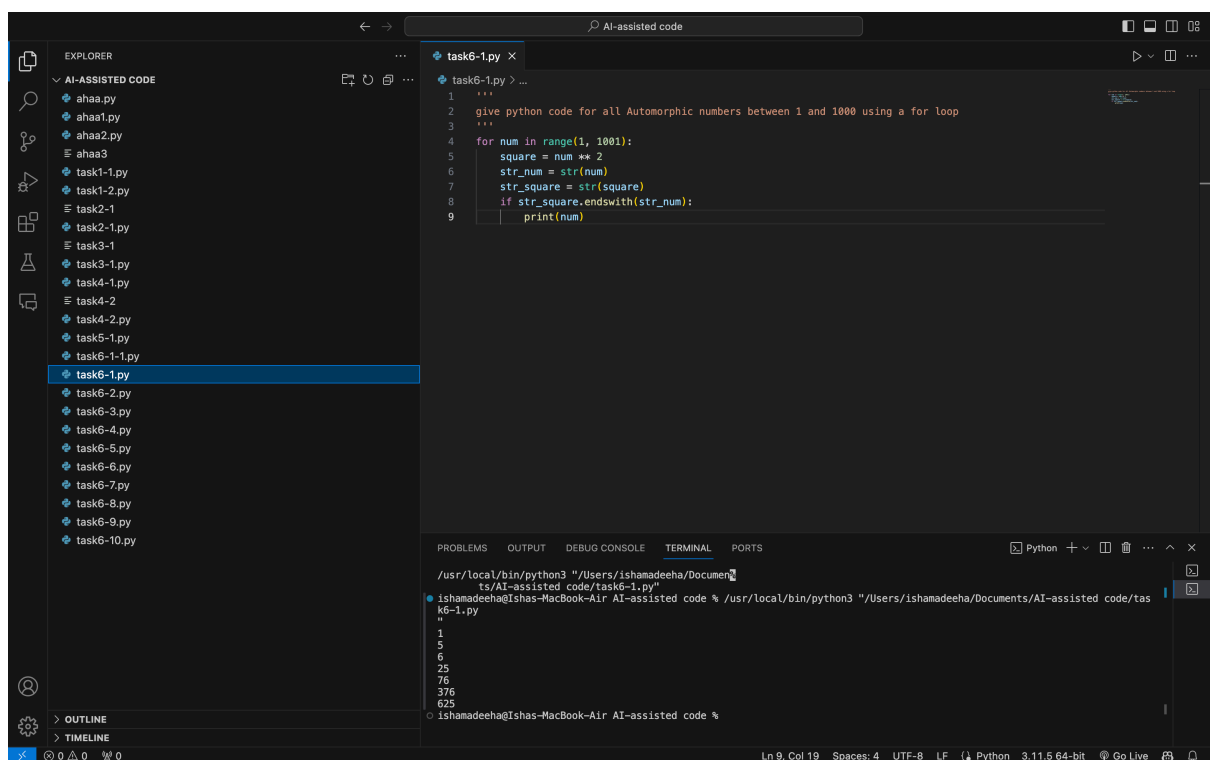
- Task: Prompt AI to generate a function that displays all Automorphic numbers between 1 and 1000 using a for loop.

- Instructions:

- o Get AI-generated code to list Automorphic numbers using a for loop.

- o Analyze the correctness and efficiency of the generated logic.

- o Ask AI to regenerate using a while loop and compare both implementations.



The screenshot shows a Visual Studio Code editor with a file explorer on the left containing a directory of files named 'task6-1.py' through 'task6-10.py'. The main editor window displays the content of 'task6-1.py', which is a Python script designed to find automorphic numbers between 1 and 1000 using a for loop. The script defines a function 'find\_automorphic\_numbers' that iterates through the range, calculates the square of each number, and checks if the square ends with the number itself. The output of the script is shown in the terminal at the bottom, listing the automorphic numbers: 1, 5, 6, 25, 76, 376, and 625.

```
task6-1.py X
1 '''
2 give python code for all Automorphic numbers between 1 and 1000 using a for loop
3 '''
4 for num in range(1, 1001):
5     square = num ** 2
6     str_num = str(num)
7     str_square = str(square)
8     if str_square.endswith(str_num):
9         print(num)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
/usr/local/bin/python3 "/Users/ishamadeeha/Documents/
ts/AI-assisted code/task6-1.py"
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
k6-1.py"
1
5
6
25
76
376
625
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

Ln 9, Col 19 Spaces: 4 UTF-8 LF Python 3.11.5 64-bit Go Live

The screenshot shows a VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with various task files. The main editor window displays a Python script in 'task6-1-1.py' that uses a while loop to find Automorphic numbers between 1 and 1000. The script checks if the square of a number ends with the number itself. The terminal at the bottom shows the output of the script, listing the Automorphic numbers: 6, 25, 76, 376, and 625.

```
1  ...
2  give python code for all Automorphic numbers between 1 and 1000 using a while loop
3  ...
4  num = 1
5  while num <= 1000:
6      square = num ** 2
7      str_num = str(num)
8      str_square = str(square)
9      if str_square.endswith(str_num):
10         print(num)
11     num += 1
12
```

Terminal Output:

```
6
25
76
376
625
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
k6-1-1.py"
```

Expected Output #1:

- Correct implementation that lists Automorphic numbers using both loop types, with explanation.

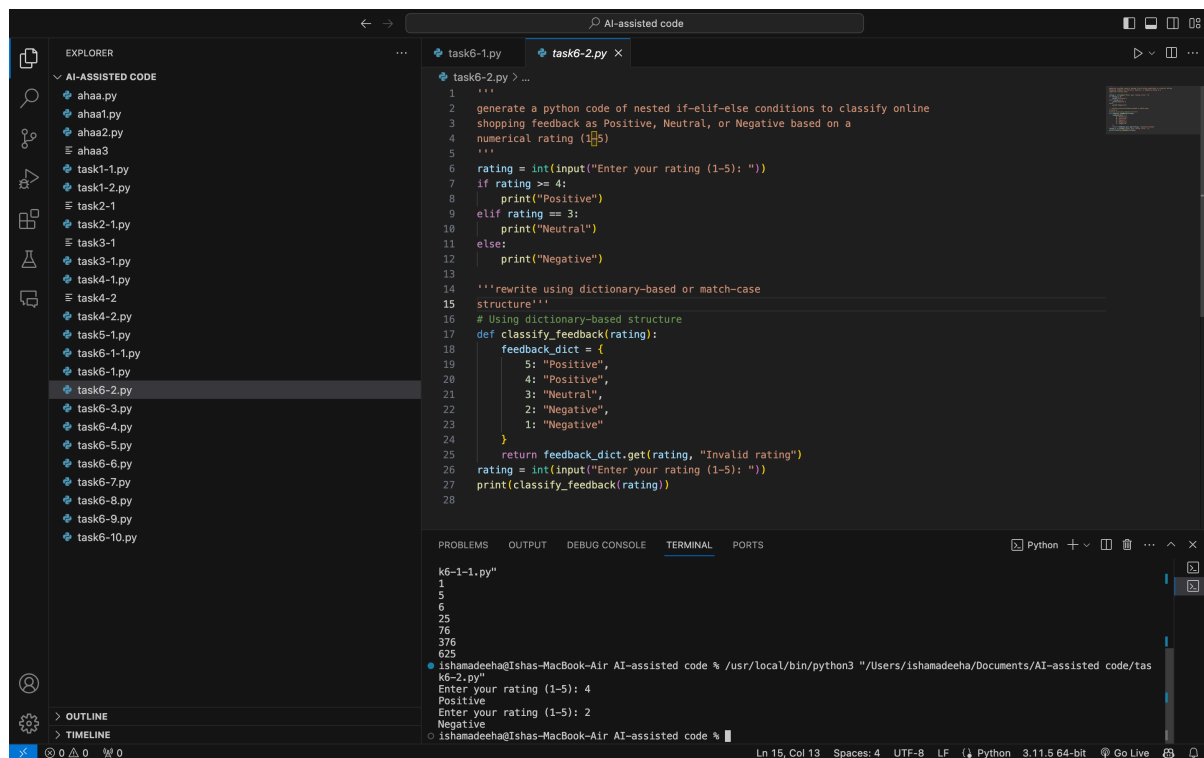
## Task Description #2 (Conditional Statements – Online Shopping Feedback Classification)

- Task: Ask AI to write nested if-elif-else conditions to classify online shopping feedback as Positive, Neutral, or Negative based on a numerical rating (1–5).
- Instructions:
  - o Generate initial code using nested if-elif-else.
  - o Analyze correctness and readability.
  - o Ask AI to rewrite using dictionary-based or match-case structure.

## Expected Output #2:

- Feedback classification function with explanation and an alternative approach.

3



```
1 '''
2 generate a python code of nested if-elif-else conditions to classify online
3 shopping feedback as Positive, Neutral, or Negative based on a
4 numerical rating (1-5)
5 '''
6 rating = int(input("Enter your rating (1-5): "))
7 if rating >= 4:
8     print("Positive")
9 elif rating == 3:
10    print("Neutral")
11 else:
12    print("Negative")
13
14 '''rewrite using dictionary-based or match-case
15 structure'''
16 # Using dictionary-based structure
17 def classify_feedback(rating):
18     feedback_dict = {
19         5: "Positive",
20         4: "Positive",
21         3: "Neutral",
22         2: "Negative",
23         1: "Negative"
24     }
25     return feedback_dict.get(rating, "Invalid rating")
26 rating = int(input("Enter your rating (1-5): "))
27 print(classify_feedback(rating))
28
```

```
k6-1-1.py"
1
5
6
25
76
376
625
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
k6-2.py"
Enter your rating (1-5): 4
Positive
Enter your rating (1-5): 2
Negative
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

## Task 3: Statistical operations

Define a function named `statistical_operations(tuple_num)` that performs the following statistical operations on a tuple of numbers:

- Minimum, Maximum
- Mean, Median, Mode
- Variance, Standard Deviation

While writing the function, observe the code suggestions provided by GitHub Copilot. Make decisions to accept, reject, or modify the suggestions based on their relevance and correctness

The screenshot shows the VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with files 'ahaa.py' through 'task6-10.py'. The main editor window displays 'task6-3.py' with the following code:

```
1 '''display a python code for function named statistical_operations(tuple_num) that performs the
2 following statistical operations on a tuple of numbers: minimum,maximum,mean, median, mode, variance, st
3 import statistics
4 def statistical_operations(tuple_num):
5     minimum = min(tuple_num)
6     maximum = max(tuple_num)
7     mean = statistics.mean(tuple_num)
8     median = statistics.median(tuple_num)
9     try:
10         mode = statistics.mode(tuple_num)
11     except statistics.StatisticsError:
12         mode = "No unique mode"
13     variance = statistics.variance(tuple_num)
14     std_deviation = statistics.stdev(tuple_num)
15
16     return {
17         "Minimum": minimum,
18         "Maximum": maximum,
19         "Mean": mean,
20         "Median": median,
21         "Mode": mode,
22         "Variance": variance,
23         "Standard Deviation": std_deviation
24     }
25
26 # Example usage
27 data = (1, 2, 2, 3, 4, 5, 5)
28 results = statistical_operations(data)
29 for key, value in results.items():
30     print(f"{key}: {value}")
```

The bottom panel shows the TERMINAL with the following output:

```
k6-2.py"
Enter your rating (1-5): 4
Positive
Enter your rating (1-5): 2
Negative
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
k6-3.py"
Minimum: 1
Maximum: 5
Mean: 3.375
Median: 3.5
Mode: 5
Variance: 2.5535714285714284
Standard Deviation: 1.5979898086569353
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

## Task 4: Teacher Profile

- Prompt: Create a class Teacher with attributes teacher\_id, name, subject, and experience. Add a method to display teacher details.
- Expected Output: Class with initializer, method, and object creation.

The screenshot shows the VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with files 'ahaa.py' through 'task6-10.py'. The main editor window displays 'task6-4.py' with the following code:

```
1 class teacher:
2     def __init__(self, name, age, subject, id, experience):
3         self.name = name
4         self.age = age
5         self.subject = subject
6         self.id = id
7         self.experience = experience
8
9     def introduce(self):
10         return f"Hello, my name is {self.name}. I am {self.age} years old and I teach {self.subject}. My
11
12 # Example usage
13 teacher1 = teacher("Alice", 30, "Mathematics", "T123", 5)
14 print(teacher1.introduce())
15 teacher2 = teacher("Bob", 45, "History", "T456", 20)
16 print(teacher2.introduce())
```

The bottom panel shows the TERMINAL with the following output:

```
Negative
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
k6-3.py"
Minimum: 1
Maximum: 5
Mean: 3.375
Median: 3.5
Mode: 5
Variance: 2.5535714285714284
Standard Deviation: 1.5979898086569353
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
k6-4.py"
Hello, my name is Alice. I am 30 years old and I teach Mathematics. My ID is T123 and I have 5 years of experience.
Hello, my name is Bob. I am 45 years old and I teach History. My ID is T456 and I have 20 years of experience.
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

## Task #5 – Zero-Shot Prompting with Conditional Validation

Use zero-shot prompting to instruct an AI tool to generate a function that validates an Indian mobile number.

### Requirements

- The function must ensure the mobile number:

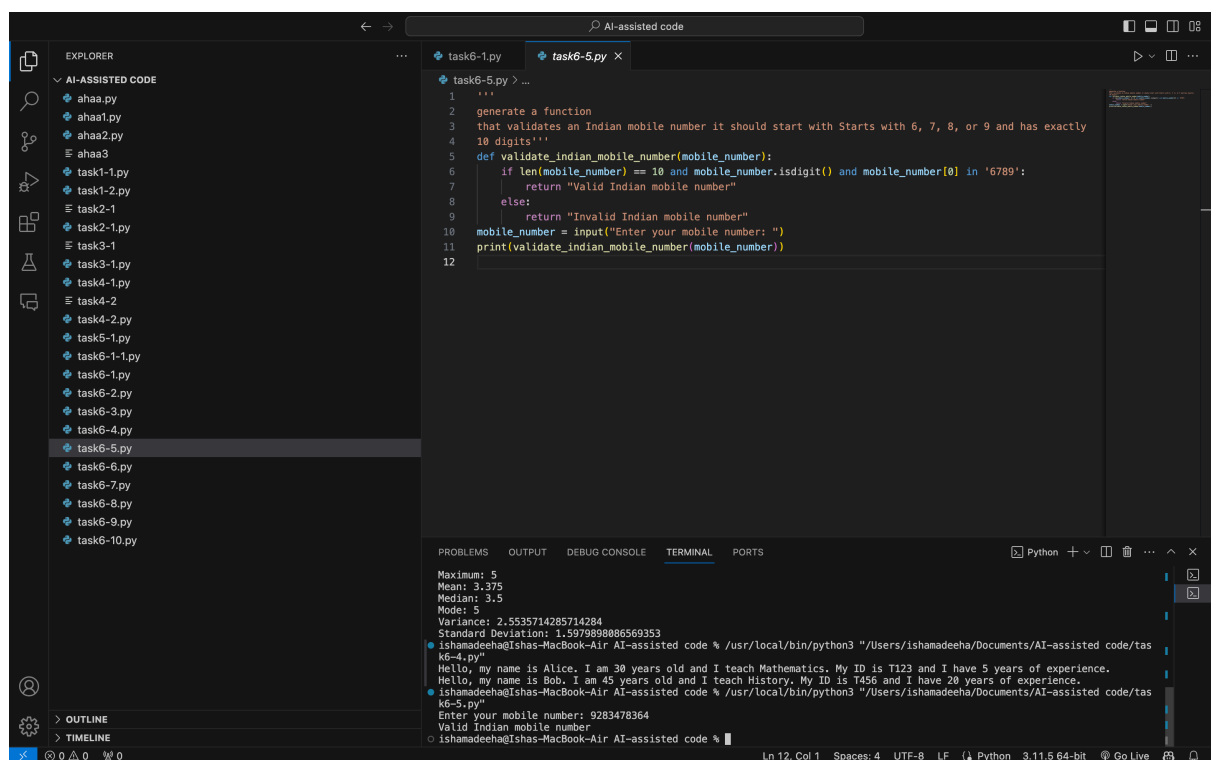
- o Starts with 6, 7, 8, or 9

- o Contains exactly 10 digits

### Expected Output

- A valid Python function that performs all required validations

without using any input-output examples in the prompt.



```
1  """
2  generate a function
3  that validates an Indian mobile number it should start with Starts with 6, 7, 8, or 9 and has exactly
4  10 digits"""
5  def validate_indian_mobile_number(mobile_number):
6      if len(mobile_number) == 10 and mobile_number.isdigit() and mobile_number[0] in '6789':
7          return "Valid Indian mobile number"
8      else:
9          return "Invalid Indian mobile number"
10     mobile_number = input("Enter your mobile number: ")
11     print(validate_indian_mobile_number(mobile_number))
12
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python + -

Maximum: 5  
Mean: 3.375  
Median: 3.5  
Mode: 5  
Variance: 2.5535714285714284  
Standard Deviation: 1.5979898086569353

- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
- kg-4.py"
- Hello, my name is Alice. I am 38 years old and I teach Mathematics. My ID is T123 and I have 5 years of experience.
- Hello, my name is Bob. I am 45 years old and I teach History. My ID is T456 and I have 20 years of experience.
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
- kg-5.py"
- Enter your mobile number: 9283478364
- Valid Indian mobile number
- ishamadeeha@Ishas-MacBook-Air AI-assisted code %

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## Task Description #6 (Loops – Armstrong Numbers in a Range)

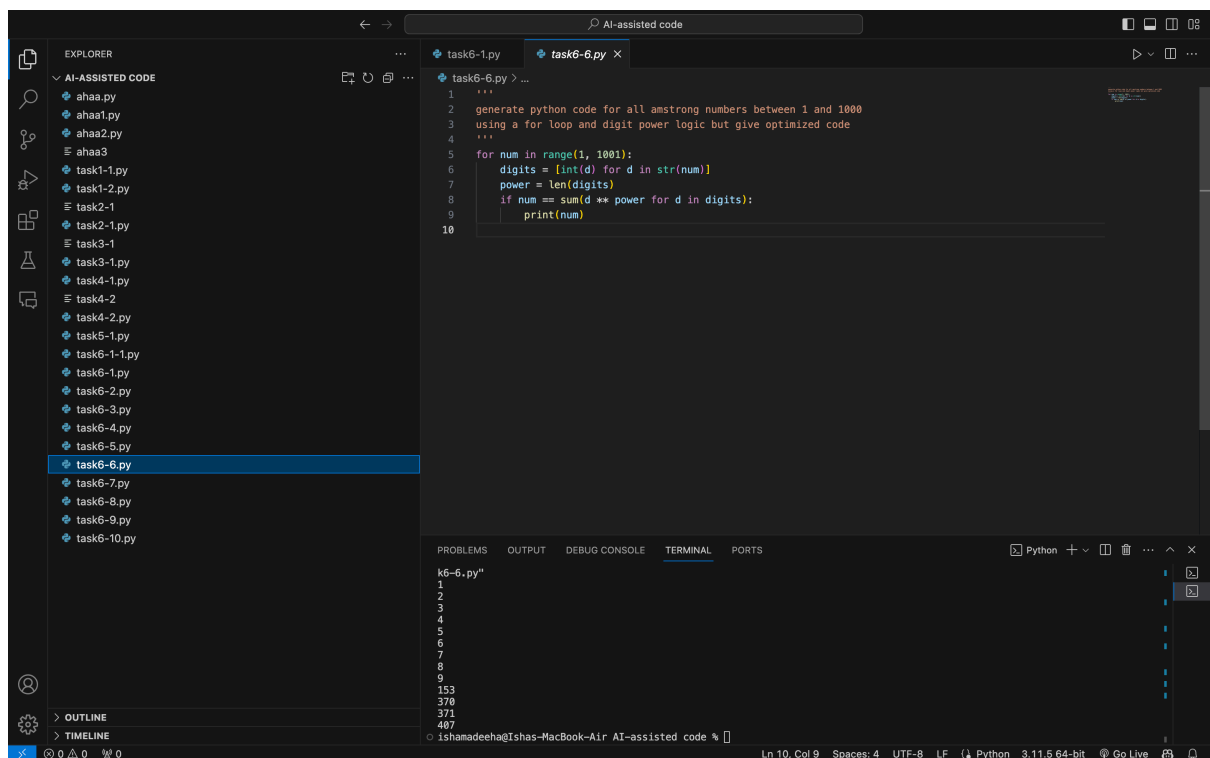
Task: Write a function using AI that finds all Armstrong numbers in a user-specified range (e.g., 1 to 1000).

Instructions:

- Use a for loop and digit power logic.
- Validate correctness by checking known Armstrong numbers (153, 370, etc.).
- Ask AI to regenerate an optimized version (using list comprehensions).

Expected Output #7:

- Python program listing Armstrong numbers in the range.
- Optimized version with explanation.



The screenshot shows a VS Code editor interface. On the left, the Explorer sidebar displays a file tree under 'AI-ASSISTED CODE' with files named 'ahaa.py', 'ahaa1.py', 'ahaa2.py', 'ahaa3', 'task1-1.py', 'task1-2.py', 'task2-1.py', 'task3-1.py', 'task4-1.py', 'task4-2.py', 'task5-1.py', 'task6-1-1.py', 'task6-1.py', 'task6-2.py', 'task6-3.py', 'task6-4.py', 'task6-5.py', 'task6-6.py' (selected), 'task6-7.py', 'task6-8.py', 'task6-9.py', and 'task6-10.py'. The main editor window shows the code for 'task6-6.py':

```
1 '''  
2 generate python code for all amstrong numbers between 1 and 1000  
3 using a for loop and digit power logic but give optimized code  
4 '''  
5 for num in range(1, 1001):  
6     digits = [int(d) for d in str(num)]  
7     power = len(digits)  
8     if num == sum(d ** power for d in digits):  
9         print(num)  
10
```

At the bottom, the TERMINAL panel shows the output of the script:

```
k6-6.py"  
1  
2  
3  
4  
5  
6  
7  
8  
9  
153  
370  
371  
407
```

The status bar at the bottom indicates the file is 'task6-6.py', the user is 'ishamadeeha@Ishas-MacBook-Air', and the Python version is '3.11.5 64-bit'.

## Task Description #7 (Loops – Happy Numbers in a Range)

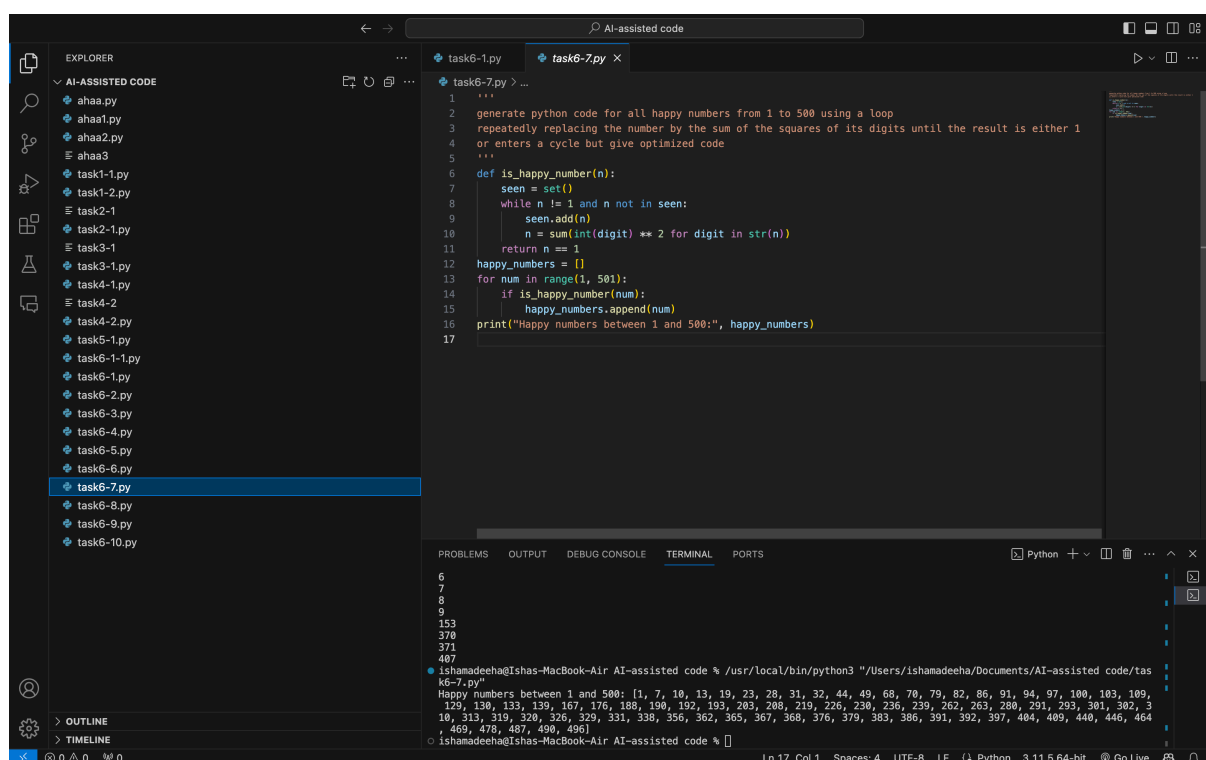
Task: Generate a function using AI that displays all Happy Numbers within a user-specified range (e.g., 1 to 500).

Instructions:

- Implement the logic using a loop: repeatedly replace a number with the sum of the squares of its digits until the result is either 1 (Happy Number) or enters a cycle (Not Happy).
- Validate correctness by checking known Happy Numbers (e.g., 1, 7, 10, 13, 19, 23, 28...).
- Ask AI to regenerate an optimized version (e.g., by using a set to detect cycles instead of infinite loops).

Expected Output #8:

- Python program that prints all Happy Numbers within a range.
- Optimized version using cycle detection with explanation.



The screenshot shows a VS Code editor with a file explorer on the left containing various task files. The main editor displays a Python script named `task6-7.py` that implements a function `is_happy_number(n)` to check if a number is happy. The script then iterates through numbers from 1 to 500, collecting all happy numbers into a list. The terminal at the bottom shows the execution output, listing the happy numbers found in the range.

```
task6-7.py
1 '''
2 generate python code for all happy numbers from 1 to 500 using a loop
3 repeatedly replacing the number by the sum of the squares of its digits until the result is either 1
4 or enters a cycle but give optimized code
5 '''
6 def is_happy_number(n):
7     seen = set()
8     while n != 1 and n not in seen:
9         seen.add(n)
10        n = sum(int(digit) ** 2 for digit in str(n))
11    return n == 1
12 happy_numbers = []
13 for num in range(1, 501):
14     if is_happy_number(num):
15         happy_numbers.append(num)
16 print("Happy numbers between 1 and 500:", happy_numbers)
17
```

Terminal Output:

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```

## Task Description #8 (Loops – Strong Numbers in a Range)

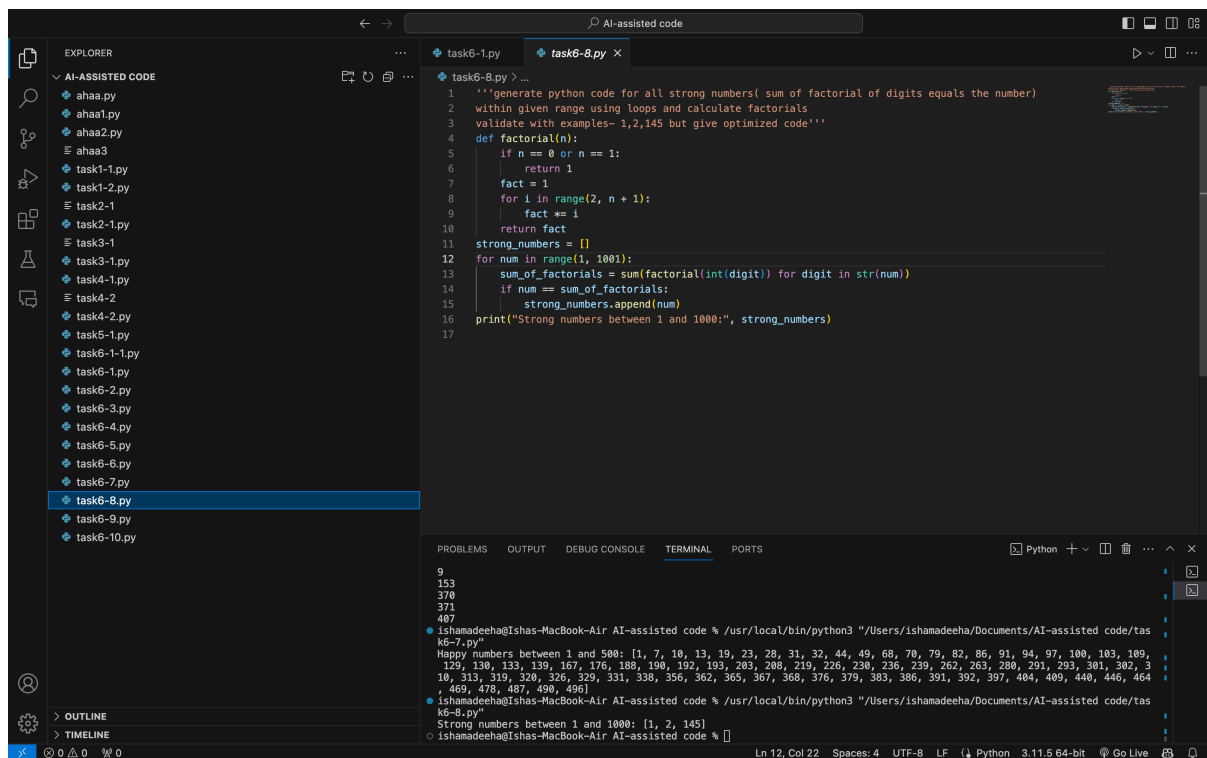
Task: Generate a function using AI that displays all Strong Numbers (sum of factorial of digits equals the number, e.g.,  $145 = 1! + 4! + 5!$ ) within a given range.

Instructions:

- Use loops to extract digits and calculate factorials.
- Validate with examples (1, 2, 145).
- Ask AI to regenerate an optimized version (precompute digit factorials).

Expected Output #9:

- Python program that lists Strong Numbers.
- Optimized version with explanation.



The screenshot shows a VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with various task files. The main editor window displays a Python script named 'task6-8.py'. The script defines a factorial function and iterates through numbers from 1 to 1000 to find strong numbers. The terminal at the bottom shows the output of the script, listing strong numbers between 1 and 1000: 1, 2, 145.

```
1 '''generate python code for all strong numbers( sum of factorial of digits equals the number)
2 within given range using loops and calculate factorials
3 validate with examples- 1,2,145 but give optimized code'''
4 def factorial(n):
5     if n == 0 or n == 1:
6         return 1
7     fact = 1
8     for i in range(2, n + 1):
9         fact *= i
10    return fact
11 strong_numbers = []
12 for num in range(1, 1001):
13     sum_of_factorials = sum(factorial(int(digit)) for digit in str(num))
14     if num == sum_of_factorials:
15         strong_numbers.append(num)
16 print("Strong numbers between 1 and 1000:", strong_numbers)
17
```

Terminal Output:

```
9
153
370
371
407
● ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task6-7.py"
Happy numbers between 1 and 500: [1, 7, 10, 13, 19, 23, 28, 31, 32, 44, 49, 68, 70, 79, 82, 86, 91, 94, 97, 100, 103, 109, 129, 130, 133, 139, 167, 176, 188, 190, 192, 193, 203, 208, 219, 226, 230, 236, 239, 262, 263, 280, 291, 293, 301, 302, 310, 313, 319, 320, 326, 329, 331, 338, 356, 362, 365, 367, 368, 376, 379, 383, 386, 391, 392, 397, 404, 409, 440, 446, 464, 469, 470, 487, 490, 496]
● ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task6-8.py"
Strong numbers between 1 and 1000: [1, 2, 145]
```



## Task #9 – Few-Shot Prompting for Nested Dictionary Extraction

### Objective

Use few-shot prompting (2-3 examples) to instruct the AI to create a function that parses a nested dictionary representing student information.

### Requirements

- The function should extract and return:

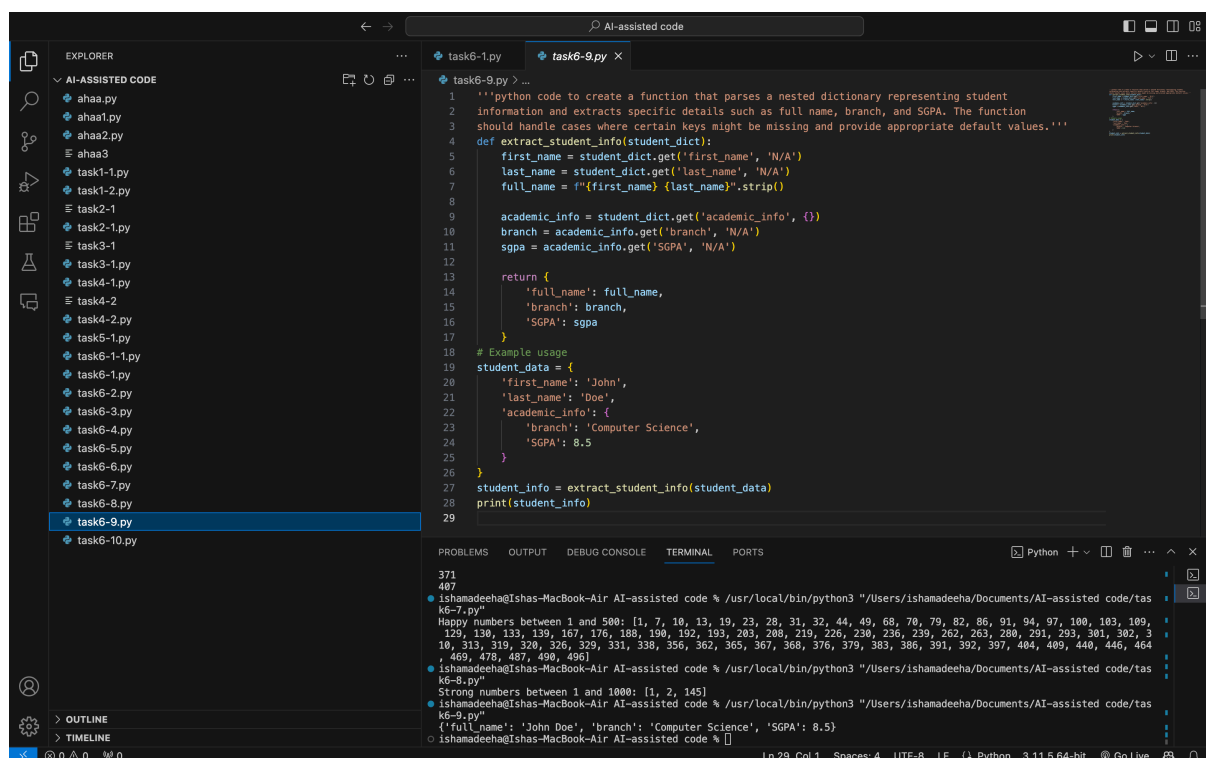
- o Full Name

- o Branch

- o SGPA

### Expected Output

A reusable Python function that correctly navigates and extracts values from nested dictionaries based on the provided examples



The screenshot shows a VS Code editor with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named 'AI-ASSISTED CODE' with several files, including 'task6-9.py' which is selected. The code editor shows the following Python code:

```
1 '''python code to create a function that parses a nested dictionary representing student
2 information and extracts specific details such as full name, branch, and SGPA. The function
3 should handle cases where certain keys might be missing and provide appropriate default values.'''
4 def extract_student_info(student_dict):
5     first_name = student_dict.get('first_name', 'N/A')
6     last_name = student_dict.get('last_name', 'N/A')
7     full_name = f'{first_name} {last_name}'.strip()
8
9     academic_info = student_dict.get('academic_info', {})
10    branch = academic_info.get('branch', 'N/A')
11    sgpa = academic_info.get('SGPA', 'N/A')
12
13    return {
14        'full_name': full_name,
15        'branch': branch,
16        'SGPA': sgpa
17    }
18
19 # Example usage
20 student_data = {
21     'first_name': 'John',
22     'last_name': 'Doe',
23     'academic_info': {
24         'branch': 'Computer Science',
25         'SGPA': 8.5
26     }
27 }
28 student_info = extract_student_info(student_data)
29 print(student_info)
```

The terminal at the bottom shows the output of the script:

```
371
407
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
k6-7.py"
Happy numbers between 1 and 500: [1, 7, 10, 13, 19, 23, 28, 31, 32, 44, 49, 68, 70, 79, 82, 86, 91, 94, 97, 100, 103, 109,
129, 130, 133, 139, 167, 176, 188, 190, 192, 193, 203, 208, 219, 226, 230, 236, 239, 262, 263, 280, 291, 293, 301, 302, 3
10, 313, 319, 320, 326, 329, 331, 338, 356, 362, 365, 367, 368, 376, 379, 383, 386, 391, 392, 397, 404, 409, 440, 446, 464
, 469, 470, 487, 490, 496]
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
k6-8.py"
Strong numbers between 1 and 1000: [1, 2, 145]
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/tas
k6-9.py"
{'full_name': 'John Doe', 'branch': 'Computer Science', 'SGPA': 8.5}
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

## Task Description #10 (Loops – Perfect Numbers in a Range)

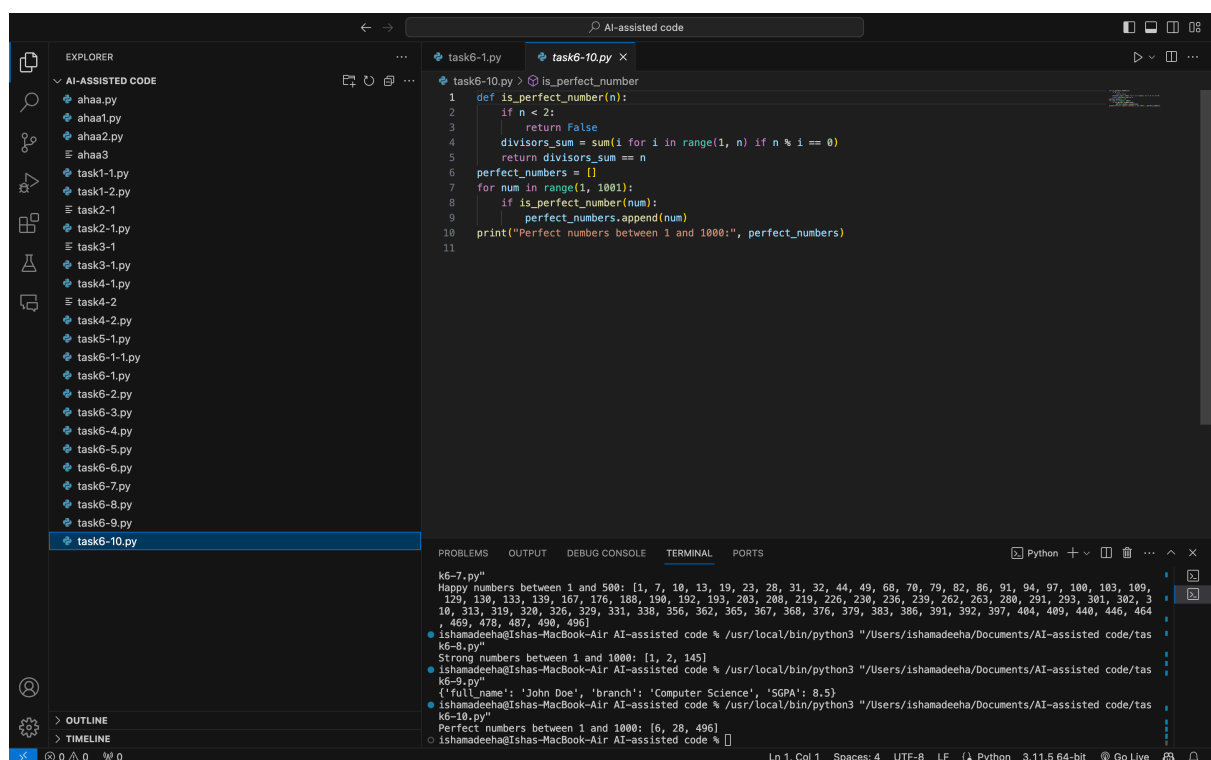
Task: Generate a function using AI that displays all Perfect Numbers within a user-specified range (e.g., 1 to 1000).

Instructions:

- A Perfect Number is a positive integer equal to the sum of its proper divisors (excluding itself).
- o Example:  $6 = 1 + 2 + 3$ ,  $28 = 1 + 2 + 4 + 7 + 14$ .
- Use a for loop to find divisors of each number in the range.
- Validate correctness with known Perfect Numbers (6, 28, 496...).
- Ask AI to regenerate an optimized version (using divisor check only up to  $\sqrt{n}$ ).

Expected Output #12:

- Python program that lists Perfect Numbers in the given range.
- Optimized version with explanation.



The screenshot shows a VS Code editor with a file explorer on the left containing various task files. The main editor displays a Python script named `task6-10.py` with the following code:

```
1 def is_perfect_number(n):
2     if n < 2:
3         return False
4     divisors_sum = sum(1 for i in range(1, n) if n % i == 0)
5     return divisors_sum == n
6
7 perfect_numbers = []
8 for num in range(1, 1001):
9     if is_perfect_number(num):
10        perfect_numbers.append(num)
11 print("Perfect numbers between 1 and 1000:", perfect_numbers)
```

The bottom panel shows the terminal output for the script:

```
K6-7.py"
Happy numbers between 1 and 500: [1, 7, 10, 13, 19, 23, 28, 31, 32, 44, 49, 68, 70, 79, 82, 86, 91, 94, 97, 100, 103, 109,
129, 130, 133, 139, 167, 176, 180, 190, 192, 193, 203, 208, 219, 226, 230, 236, 239, 262, 263, 280, 291, 293, 301, 302, 3
10, 313, 319, 320, 326, 329, 331, 338, 356, 362, 365, 367, 368, 376, 379, 383, 386, 391, 392, 397, 404, 409, 440, 446, 464
, 469, 478, 487, 490, 496]
K6-8.py"
Strong numbers between 1 and 1000: [1, 2, 145]
K6-9.py"
{'full_name': 'John Doe', 'branch': 'Computer Science', 'GPA': 8.5}
K6-10.py"
Perfect numbers between 1 and 1000: [6, 28, 496]
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

